Steven Festenstein - Staff Software Engineer

Syracuse, NY

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Overview

Staff Software Engineer with 30 years of experience designing high-performance applications using C++, Qt, Java, and Python. Specializes in SOLID object-oriented programming, GUI development, and passive Electronic Warfare. Experience with Real-Time Operating Systems (RTOS), embedded development, and distributed systems, with a proven track record of delivering quality software solutions and leading technical training for engineering teams.

Core Skills

- Expert in SOLID object-oriented programming with C++.
- Skilled in GUI development using C++ and QML with the Qt Framework.
- GCC, Clang, CMake, QMake, and GNU Make for cross-platform development.
- Distributed systems design with ZeroMQ, gRPC, and DDS.
- AI-Assisted Software Development.
- Long history of working on RF systems.
- Embedded development with VxWorks, LynxOS, and PetaLinux.
- Java development with expertise in JavaFX and Swing for GUI applications.
- .NET, C#, and Python for diverse software solutions.
- Scrum-based Agile development methodologies.
- Technical Trainer: developed and presented training modules to large audiences.

Experience

Staff Software Engineer, Lockheed Martin - Syracuse, October 1996 to Present

AI/ML Engineer at the Lockheed Martin Artificial Intelligence Center (LAIC) - March 2026 to Present

Co-developer for the StarOS C++ Software Development Kit. Designing and deploying plugin-style libraries for multiple platforms. Employing modern standards and development processes. Leveraged containerized development and AI coding assistants to maximize quality and productivity.

SEWIP Block 2 - April 2025 to May 2026

Developed HMIs by porting legacy Java code to C++/Qt. Modernized the build process using CMake and updated developer tools to industry standards. Also updated the embedded backend in VxWorks to assist with calibration and report system health.

Terrestrial Layer System (TLS) UI Team - April 2024 to April 2025

Implemented a geographic map application to paint tracks on top of a map that can be panned and zoomed, much like Google Maps. Updated build process to use CMake. Updated internal data models to use relational databases (SQLite and MySQL). Addressed software defects, stability, and performance.

Shipboard Panoramic Electro-Optical Infrared (SPEIR) - HMI Lead - March 2023 to April 2024

Led a team of 4 engineers to develop front-end software for the SPEIR program, using Qt Framework and C++ on RHEL 8. Employed video streaming technology to display multiple videos from various imaging sensors on naval surface vessels. Developed an OpenGL Video Player to render streaming videos in H.264 and H.265 formats. Developed middleware using ZeroMQ and Google Protocol Buffers to interact with backend.

BLQ-10 (Subsurface Electronic Warfare) - 2015 to 2023

OMI Working Group Software Lead; developed next-generation user interfaces from scratch. Took responsibility for all aspects of software development. Led a team to implement software features and updates under demanding timelines. Employed OpenGL and the Qt framework with C++ to create an attractive and responsive user interface. Prepared software for customer demonstration and feedback every 6-8 weeks.

Technical Interviewer - 2014 to Present

- Conducted hundreds of interviews for software roles, including interns, experienced professionals, and contractors.
- Designed coding exercises and technical questions to streamline the interview process.
- Significantly reduced employee turnover.

Technical Trainer - 2018 to Present

- Developed and delivered a nine-part C++ curriculum for new software engineers. Conducted the curriculum in front of hundreds of participants.
- Created a two-part course on OpenGL and Qt Framework, enhancing team capabilities in GUI development.
- Presented advanced C++ lectures: "CMake Project Management" and "Distributed Computing with ZeroMQ."

Further Experience on Request

Education

- **Bachelor of Science in Physics (Mathematical Physics and Astrophysics)**
  State University of New York at Fredonia *[1991-1994]*
- **Bachelor of Science in Electrical Engineering (Signal Processing)**
  State University of New York at Buffalo *[1994-1996]*

Personal Software Projects

Radio Wizard - github.com/festensoft/RadioWizard Linux and macOS SDR application in the style of GQRX, but more attractive and easier to use.

Omniscope DDS - github.com/festensoft/OmniscopeDDS Linux and macOS application to snoop and debug DDS traffic. C++ backend with HTML/CSS frontend.

Football Squares - football-squares.com AI-Assisted web application using React. Implements a novel football (Super Bowl) board.